

Oliver Zhang

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Summary

I'm a versatile mechanical engineer with extensive experience designing and building complex electro-mechanical systems. I thrive in highly collaborative interdisciplinary teams, and I love finding elegant solutions that bring innovative ideas to life.

I'm an Australian citizen.

Education

Carnegie Mellon University, BS in Mechanical Engineering with an additional major in Robotics with University Honors – Pittsburgh, USA Aug 2015 – May 2019

- Completed a wide range of course work from mechanical engineering fundamentals to computer vision, artificial intelligence, and systems engineering.
- Worked with cross-disciplinary teams on multiple projects, including building an amphibious vehicle, a self-navigating drone, and an autonomous delivery robot.

Experience

Senior Mechanical Engineer, PA Consulting – San Francisco, USA July 2019 – present

Collaborating with diverse engineers and designers to find the path through complex challenges and turn inventive ideas into innovative products.

- The team's go-to expert for robotics, systems engineering, and analysis.
- Delivering projects across many different industries, from medical robotics to industrial food production.
- Brainstorming ideas and developing concepts to find the best path forward.
- Building prototypes at the right fidelity to rapidly drive to the optimal design.
- Architecting complex systems by breaking them down into key interactions and applying first principles analysis and modelling.
- Collaborating with a wide range of clients from agile tech startups to established global leaders.
- Coordinating highly interdisciplinary teams spread across multiple geographies, such as electrical engineers, software engineers, and industrial designers.
- Growing the team's capabilities through weekly forums and training sessions, covering topics like analysis, sketching, or programming.

Project Engineer Intern, Autel Robotics – Pittsburgh, USA May 2017 – Aug 2017

Implemented tracking and gesture recognition algorithms for embedded execution on a quadcopter.

Skills

Programming: Python, C/C++, MATLAB, ROS, Unity

Mechanical Engineering: Technology development and prototyping, Analysis of complex systems, Electromechanical systems, Design for manufacture, SolidWorks, GD&T, Tolerance stack analysis

Systems Engineering: Requirements development, System architecture design, Risk management

Languages: English (native speaker), Mandarin (native speaker)

Patents

US11058204B2 July 2021

Automated total nail care systems, devices, and methods

Projects

Novel Electromechanical Machine

Aug 2025 – Feb 2026

Developing a consumer facing electromechanical machine for a global technology company.

- Owned design of an Alpha prototype system with 7 actuated degrees of freedom.
- Designed complex motion systems in heavily space constrained and finish constrained environment.
- Balanced strict industrial design intent with reliability, while meeting aggressive schedule.
- Interfaced with a diverse team of electrical engineers, software engineers, and industrial designers.
- Led prototype build effort, coordinating integration of PLC electronics, wiring of motors and sensors, and assembly of complex mechanical subsystems.
- Final prototype's reliability and quality exceeded expectations and allowed client to fully de-risk intended user experience with company executives.

Novel Cable Splicing Machine

Aug 2024 – Aug 2025

Developing an underground cable splicing machine that automates a complex process typically done by a trained technician.

- Built complex system requirements drawn from multiple stakeholders, and managed key risks.
- Designed system architecture to maximise reliability and meet stringent weight and space constraints.
- Developed multiple electromechanical mechanisms that each perform dexterous operations that previously required highly coordinated human actions.

Novel Smart Fitness Machine

Aug 2022 – Aug 2024

Developed a novel fitness machine that is controllable wirelessly and can report data from multiple integrated sensors.

- Implemented and refined controls algorithms to realistically simulate inertia with an electric motor.
- Built dynamometer rig with software that simulates human motion and displays metrics in real-time.
- Developed system requirements, managing traceability and designing test plans.
- Led verification and validation of the prototypes across multiple locations globally.
- Coordinated a highly cross-functional team with industrial designers, electrical engineers, firmware engineers, and mechanical engineers.
- Delivered high quality prototypes that demonstrated best-in-class user experience and performance, standing up to rigorous testing by world-class athletes.

Optimising Industrial Food Production

Aug 2023 – Nov 2023

Analysed an industrial food production line to identify root causes of defects and improve throughput without changing line footprint.

- Analytically modelled complex dynamic behavior of key components in the production line leveraging high-speed camera footage collected on-site.
- Developed several concepts for improvement, covering the full solution space.
- Client implemented designs and achieved 47% increased throughput without increasing line footprint.

Optimising Commercial Appliance

Mar 2022 – Nov 2022

Analysed airflow through a commercial appliance to optimise efficiency.

- Built thermodynamic simulation of the system based on lab data collected from dozens of sensors.
- Developed a feedback control system that dynamically adjusts air flow rate for best system performance.
- Coordinated with firmware engineers and technicians to deploy and test control system.

Electromechanical Security Product

Feb 2022 – Mar 2022

Rapidly developed concepts and prototypes for addressing a critical issue in a electromechanical security product.

- Analysed the existing design, identifying root causes of issues found in the field.
- Brainstormed ideas and mapped the solution space with a dozen key architectures.
- Built multiple prototypes that successfully demonstrated five leading concepts.

Medical Robot Arm

Apr 2021 – Feb 2022

Developed mechanical designs for one of the arms in a medical robot.

- Optimised structural components with stringent weight and stiffness requirements.
- Analysed and designed parts with extremely tight tolerances to ensure assemblability while minimising clearance.
- Built comprehensive analytical model of a linkage mechanism to predict motion, loads, and deflection.

Novel Subsystem Prototype

Sept 2020 – Apr 2021

Developed a proof-of-principle prototype to de-risk a key robotic subsystem.

- Developed a complex analytical model of the behavior of elastic materials fed from a spool.
- Experimented with medical grade extruded tube materials, coatings, and cross-sectional profiles to generate desired properties such as bending stiffness, resilience, creep, and friction.
- Designed and built a robotic system that process various sensor inputs to drive a custom control algorithm. The system uses clinical trial data to simulate real user motion, allowing high fidelity testing of expected behavior.

Evolv Express® Threat Detection System

Nov 2019 – Sept 2020

Developed manufacturable prototype in support of client electronics, addressing challenging structural, materials, mobility, and environmental requirements.

- Architected mechanical structure to improve stiffness, assemblability, and serviceability.
- Converted fibreglass sheet-based design into structural foam/RIM/plastic extrusions.
- Performed comprehensive Six sigma tolerance stack analysis and implemented GD&T tolerance controls.
- Managed complex assembly with master modelling techniques.
- Final prototypes secured customers and funding, eventually leading to public listing of client's company.

Novel Chemical Treatment Prototype

Aug 2019 – Nov 2019

Developed a prototype machine for temperature-controlled chemical treatment

- Developed Peltier cooling channels that maintain the temperature of a chemical being cycled through the system.
- Designed sound dampening features to reduce the noise impact from vacuum pumps.
- Built a portable sheet metal enclosure to hold sensitive chemicals and electronics.
- Final prototype successfully demonstrated the chemical treatment technology, allowing the client to secure funding for further development.

Novel Human Machine Interface (HARMAN International)

May 2018 – Aug 2018

Developed shape-shifting device to support client investigations into novel human-machine interfaces.

- Developed concepts through brainstorming, cross-pollination, and down-selection processes.",
- Built an integrated electromechanical prototype that was displayed at a major international trade show.",
- Built a graphics user interface to prove prototype usability and reliability.",
- Final prototype successfully demonstrated at a major international trade show.